

summarisation, and SW3 for a custom AI task, such as solving mathematical problems.

The system operates with a 5V supply powering the ESP32-CAM core, while the 3.3V line supports the peripherals, including the OLED, buzzer, and push buttons.

Software

The code is developed using the Arduino IDE. In the initial section, libraries for Wi-Fi configuration, the ESP32-CAM module, the SH1106 OLED display, and the HTTP client are included. Wi-Fi credentials, including the SSID and password, are defined in the code, along with the OpenAI API key for AI-powered image analysis.

To begin, an OpenAI API key needs to be generated at <https://platform.openai.com/api-keys>. After logging in to an OpenAI account, select 'Create API key' (create a new secret key). Assign a name to the key and proceed to make it. The key must be saved securely, as it will be required later in the code. Fig. 3 shows the OpenAI API dashboard with a generated secret key and its details.

Next, the U8g2lib library and other essential libraries should be installed in the Arduino IDE. The Wi-Fi SSID and password must then be configured in the code, followed by the API key created and saved earlier. Fig. 4 shows a screenshot of the OpenAI API key creation window, which includes instructions on how to securely save the key, along with permissions for reading and writing API resources.

In the next part of the code, the setup process is managed by the initialiseSystem function, which establishes Wi-Fi connectivity and configures pin modes for three push buttons and a buzzer. Camera initialisation is handled by the initialiseCamera function, which sets parameters such as resolution and image quality.

Two utility functions enhance

the user experience: the showText function displays text on the OLED screen while accounting for height, width, and font constraints, and the playSound function provides audio feedback through the buzzer.

The main loop is implemented in the handleButtons function,

which continuously monitors button presses. Each button is mapped to a customisable prompt for tasks such as solving mathematical problems, language translation, or generating image summaries. Fig. 5 shows the code snippet configuring the API key and Wi-Fi credentials.

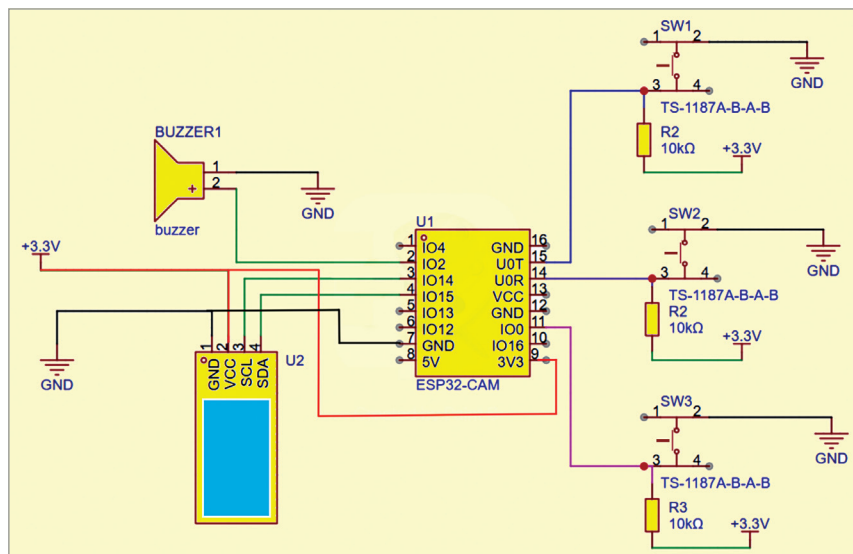


Fig. 2: Circuit diagram

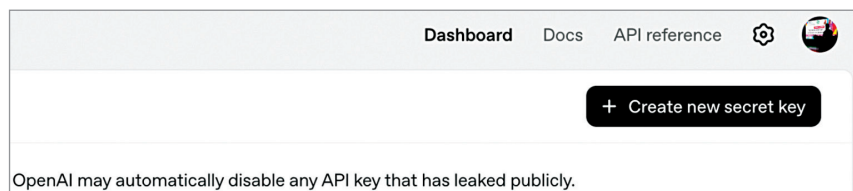


Fig. 3: OpenAI API dashboard

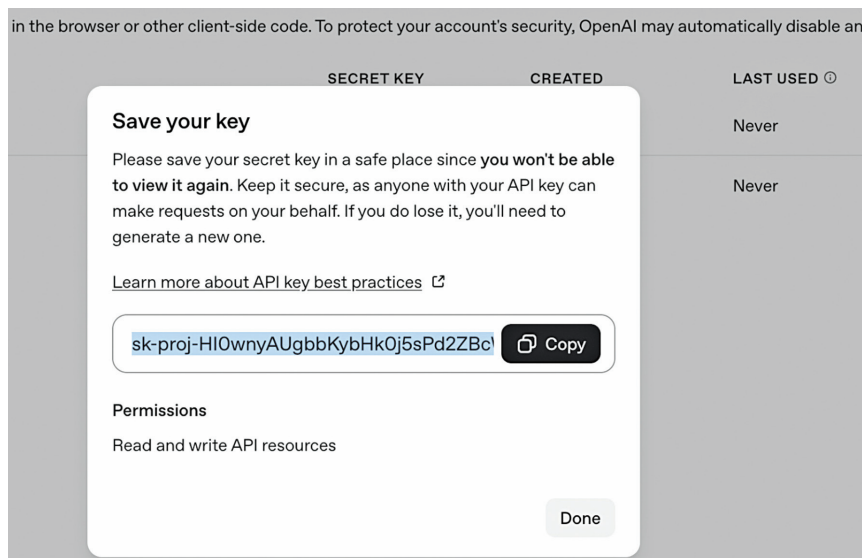


Fig. 4: Screenshot of the OpenAI API key creation